DPO and RLHF

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Bradley-Terry Model

We are given an LLM model $\pi_{ref}(y|x)$, where x is the prompt and y is the output. If the output consists of H tokens, then

$$\pi_{ref}(y|x) = p_{ref}(y_0|x)\, p_{ref}(y_1|x, y_0^1) \cdots p_{ref}(y_{H-1}|x, y_0^{H-2})$$

We assume that there is a model π_{ref} which is trained from all the data on the internet. We now want to fine-tune the model to get a new π that is aligned with human preferences while not straying from π_{ref}. To do this, we assume that humans give a score of $r(x,y)$ for output y given a prompt x. We also assume x is generated by some distribution p_x. Thus, the goal is to

$$\max_{\pi} E_{x \sim p_x, y \sim \pi(y|x)}\left[r(x,y) - \beta D_{KL}\left(\pi(\cdot|x) \| \pi_{ref}(\cdot|x)\right)\right],$$

$$= \max_{\pi} \sum_{x,y} r(x,y)\, p_x(x)\, \pi(y|x) - \beta \sum_x p_x(x) \sum_y \pi(y|x) \log \frac{\pi(y|x)}{\pi_{ref}(y|x)}$$

Differentiating w.r.t. $\pi(y|x)$ (need to be careful here since $\sum_y \pi(y|x) = 1$, $\pi(y|x) \geq 0$)

$$r(x,y)\cancel{p_x(x)} - \beta \cancel{p_x(x)}\left[\log \frac{\pi(y|x)}{\pi_{ref}(y|x)} - 1\right] = 0$$

$$\log \frac{\pi(y|x)}{\pi_{ref}(y|x)} = \frac{1}{\beta} r(x,y) + 1$$

$$\Rightarrow \boxed{\pi(y|x) = \frac{\pi_{ref}(y|x)\, e^{\frac{1}{\beta} r(x,y)}}{Z(x)},} \quad — Ⓐ$$

where $Z(x)$ is a normalizing constant. But we don't know r!! So, this solution cannot be implemented. Further, $Z(x)$ is computationally infeasible to compute.

First, let's try to think about how to estimate r. We do this using human feedback. Suppose given a prompt x, we provide two possible outputs y_1, y_2 to a human, we assume that that the human would prefer y_1 over y_2 according to the following Bradley-Terry model:

$$P(y_1 \succ y_2 | x) = \frac{e^{r(x,y_1)}}{e^{r(x,y_1)} + e^{r(x,y_2)}}$$

$$= \frac{1}{1 + e^{-(r(x,y_1) - r(x,y_2))}}$$

$$=: \sigma\left(r(x,y_1) - r(x,y_2)\right)$$

Suppose $x \sim p_x$, $y_1, y_2 | x \sim \pi_{ref}(\cdot|x)$ and human preference between y_1, y_2 denoted by y_w, y_ℓ (for winner & loser). Then, the ML estimator of r is given by

$$\max_r \sum_{i=1}^{N} \log\left(\sigma\left(r(x_i, y_{wi}) - r(x_i, y_{\ell i})\right)\right),$$

assuming that there are N samples.

From Ⓐ,

$$r(x,y) = \beta \log \frac{\pi(y|x)\, Z(x)}{\pi_{ref}(y|x)}$$

$$\Rightarrow r(x, y_w) - r(x, y_\ell) = \beta \log \frac{\pi(y_w|x)}{\pi_{ref}(y_w|x)} \cdot \frac{\pi_{ref}(y_\ell|x)}{\pi(y_\ell|x)}$$

(Interestingly $Z(x)$ has vanished)

Thus, the ML estimator is

$$\max_{\pi} \sum_{i=1}^{N} \log\left(\sigma\left(\beta \log \frac{\pi(y_{wi}|x)}{\pi_{ref}(y_{wi}|x)} \cdot \frac{\pi_{ref}(y_{\ell i}|x)}{\pi(y_{\ell i}|x)}\right)\right)$$

The LLM has training parameters θ. In terms of this, we have

$$\boxed{\max_{\theta} \sum_{i=1}^{N} \log\left(\sigma\left(\beta \log \frac{\pi_\theta(y_{wi}|x)}{\pi_{ref}(y_{wi}|x)} \frac{\pi_{ref}(y_{\ell i}|x)}{\pi_\theta(y_{\ell i}|x)}\right)\right)}$$

→ A non-convex problem, but one that we assume can be solved.

RLHF: In contrast to DPO, RLHF first assumes a parametric form $r_\phi(x,y)$ for r and computes ϕ by solving the ML problem:

$$\max_{\phi} \sum_{i=1}^{N} \log\left(\sigma\left(r_\phi(x_i, y_{wi}) - r_\phi(x_i, y_{\ell i})\right)\right),$$

Once this is solved, we have a reward model and we can then try to solve

$$\max_{\theta} E_{x \sim p_x, y \sim \pi_\theta(y|x)}\left[r_\phi(x,y) - \beta D_{KL}\left(\pi_\theta(\cdot|x) \| \pi_{ref}(\cdot|x)\right)\right].$$

As we saw before, $Z(x)$ cannot be calculated, so it is hard to solve the problem in closed-form. It is solved using RL methods such as PPO. To see how to apply PPO, note that the objective can be written as

$$\sum_x p_x(x) \sum_y \pi_\theta(y|x)\left[r_\phi(x,y) - \beta \underbrace{\sum_y \pi_\theta(y|x) \log \prod_{h=0}^{H-1} \frac{\pi_\theta(y_h|x, y_0^{h-1})}{\pi_{ref}(y_h|x, y_0^{h-1})}}_{\sum_y \pi_\theta(y|x) \sum_{h=0}^{H-1} \log \frac{\pi_\theta(y_h|x, y_0^{h-1})}{\pi_{ref}(y_h|x, y_0^{h-1})}}\right]$$

$$= \sum_x p_x(x) \sum_y \pi_\theta(y|x)\left[r_\phi(x,y) - \beta \sum_{h=0}^{H-1} \log \frac{\pi_\theta(y_h|x, y_0^{h-1})}{\pi_{ref}(y_h|x, y_0^{h-1})}\right]$$

So, you can think of the problem as a horizon H MDP with initial state chosen by p_x, policy is π_θ, the state transition is deterministic and the reward is

$$-\beta \log \frac{\pi_\theta(y_h|x, y_0^{h-1})}{\pi_{ref}(y_h|x, y_0^{h-1})} \quad \text{if } h \leq H-2$$

$$r_\phi(x,y) - \beta \log \frac{\pi_\theta(y_h|y_0^{h-1})}{\pi_{ref}(y_h|x, y_0^{h-1})} \quad \text{if } h = H-1$$

The optimal θ is calculated using PPO for this finite horizon problem.